# Database Systems and Web (15B11CI312)

# Database Systems and Web

Lecture 10: EER

# Contents to be covered

- Enhanced ER Modelling
- Subclass
- Superclass
- Generalization
- Specialization

# The Enhanced Entity-Relationship (EER) Model

**Enhanced ER (EER) model**

- Created to design more accurate database schemas
  - Reflect the data properties and constraints more precisely
- More complex requirements than traditional applications

# Subclasses, Superclasses, and Inheritance

EER model includes all modeling concepts of the ER model

In addition, EER includes:

- **Subclasses** and **superclasses**
- **Specialization** and **generalization**
- **Category** or **union type**
- **Attribute** and **relationship inheritance**

# Subclasses and Superclasses

An entity type may have additional meaningful subgroupings of its entities

- Example: EMPLOYEE may be further grouped into:
  - SECRETARY, ENGINEER, TECHNICIAN, …
    - Based on the EMPLOYEE's Job
  - MANAGER
    - EMPLOYEEs who are managers
  - SALARIED_EMPLOYEE, HOURLY_EMPLOYEE
    - Based on the EMPLOYEE's method of pay

EER diagrams extend ER diagrams to represent these additional subgroupings, called *subclasses* or *subtypes*

# Subclasses and Superclasses

Each of these subgroupings is a subset of EMPLOYEE entities

Each is called a subclass of EMPLOYEE

EMPLOYEE is the superclass for each of these subclasses

These are called superclass/subclass relationships:

- EMPLOYEE/SECRETARY
- EMPLOYEE/TECHNICIAN
- EMPLOYEE/MANAGER
- …

# Subclasses and Superclasses

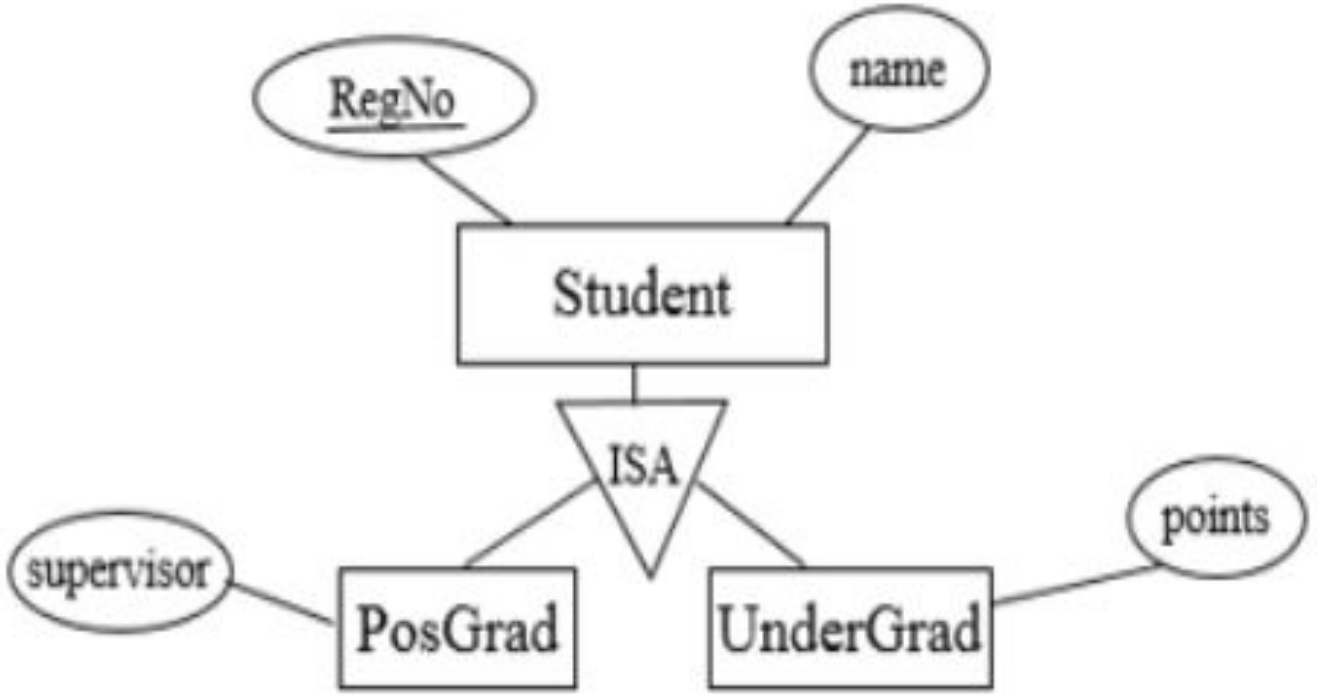
These are also called **IS-A relationships**

- SECRETARY IS-A EMPLOYEE, TECHNICIAN IS-A EMPLOYEE, ….

**Note:** An entity that is member of a subclass represents the same real-world entity as some member of the superclass:

- **The subclass member is the same entity in a *distinct specific role***
- An entity cannot exist in the database merely by being a member of a subclass; it must also be a member of the superclass
- A member of the superclass can be optionally included as a member of any number of its subclasses

# IS-A Relationship



# Attribute Inheritance in Superclass / Subclass Relationships

An entity that is member of a subclass *inherits*

- All attributes of the entity as a member of the superclass
- All relationships of the entity as a member of the superclass

Example:

- In the previous slide, SECRETARY (as well as TECHNICIAN and ENGINEER) inherit the attributes Name, SSN, …, from EMPLOYEE
- Every SECRETARY entity will have values for the inherited attributes

# Membership Constraints

**Predicate defined subclasses**

- The subclass is defined through a predicate on the attributes of the superclass
- Display a predicate-defined subclass by writing the predicate condition next to the line attaching the subclass to its superclass

**Attribute defined subclasses**

- The subclasses in the specialization are all defined by the same attribute of the superclass
  - Attribute is called the defining attribute of the specialization
  - Example: JobType is the defining attribute of the specialization {SECRETARY, TECHNICIAN, ENGINEER} of EMPLOYEE

**User defined subclasses**

- Membership in the subclasses is determined at the insertion operation level

# Specialization and Generalization

# Specialization

Specialization is the process of defining a set of subclasses of a superclass

The set of subclasses is based upon some distinguishing characteristics of the entities in the superclass

- Example: {SECRETARY, ENGINEER, TECHNICIAN} is a specialization of EMPLOYEE based upon *job type*.
  - May have several specializations of the same superclass

Subclass can define:

- **Specific attributes**
- **Specific relationship types**

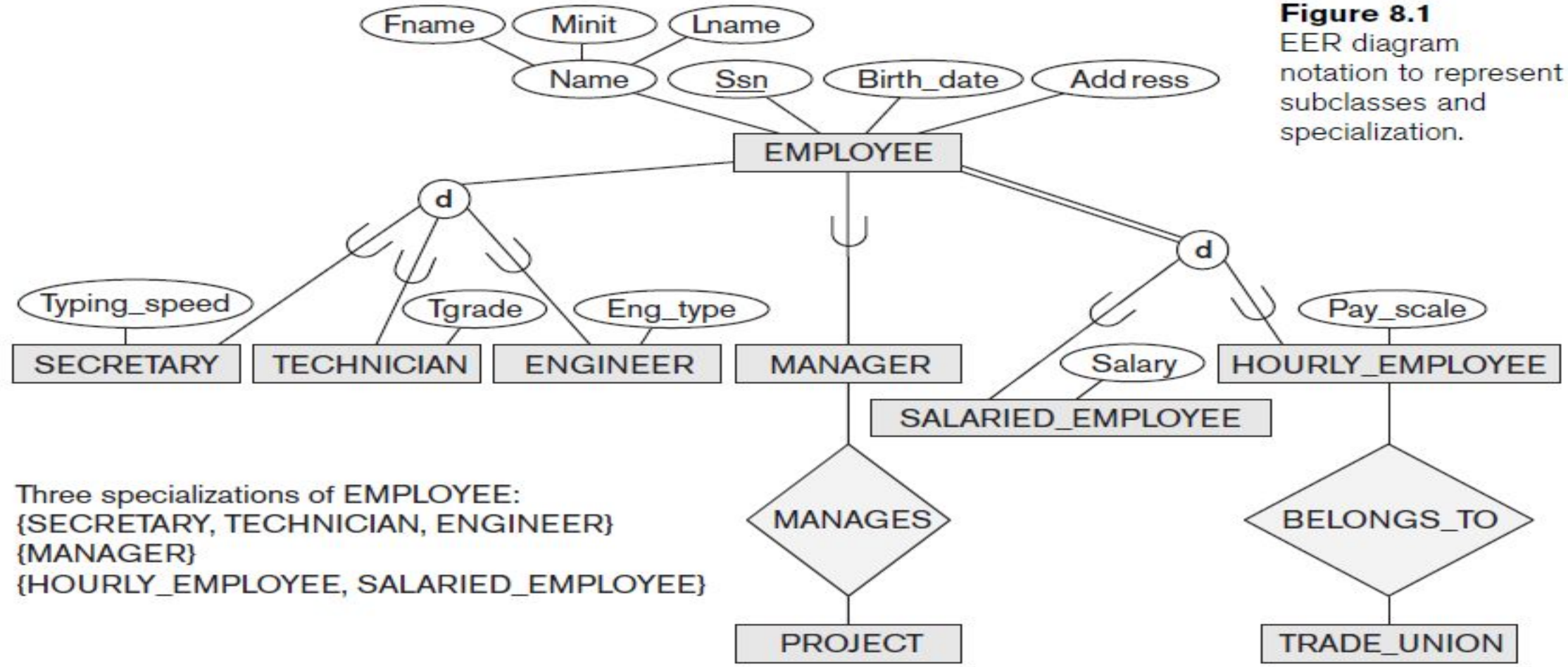



**Figure 8.1**
EER diagram notation to represent subclasses and specialization.

*Source: Fundamentals of database systems / Ramez Elmasri, Shamkant B. Navathe.—6th ed, Pearson Publications*

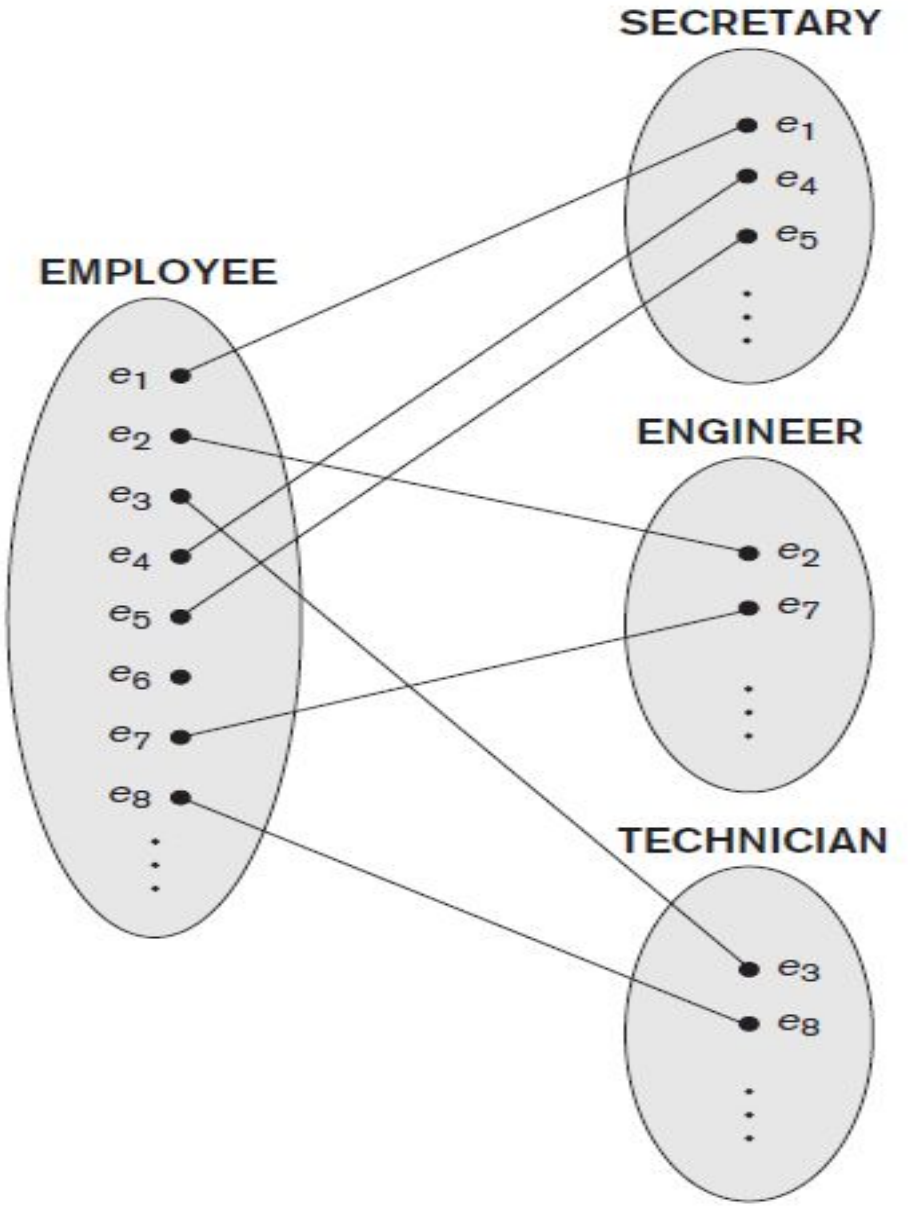



**Figure 8.2**
Instances of a specialization.

*Source: Fundamentals of database systems / Ramez Elmasri, Shamkant B. Navathe.—6th ed, Pearson Publications*

# Generalization

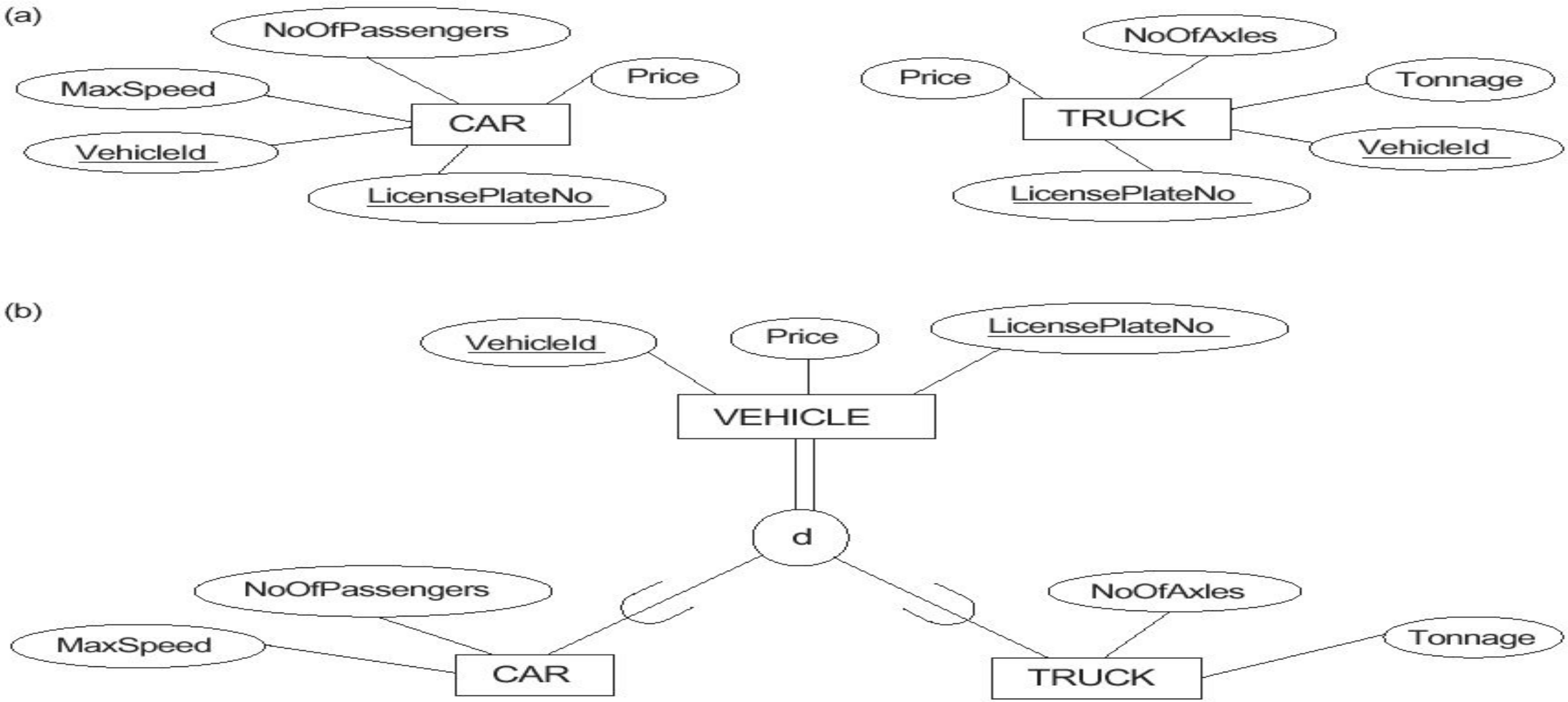
**Generalize** into a single **superclass**

- Original entity types are special subclasses

**Generalization**

- Process of defining a generalized entity type from the given entity types

**Figure 4.3** Examples of generalization. (a) Two entity types CAR and TRUCK. (b) Generalizing CAR and TRUCK into VEHICLE.




*Source: Fundamentals of database systems / Ramez Elmasri, Shamkant B. Navathe.—6th ed, Pearson Publications*

# Constraints on Specialization and Generalization

Two basic constraints can apply to a specialization/generalization:

- Disjointness Constraint:
- Completeness Constraint:

# Disjointness Constraints

Specifies that the subclasses of the specialization must be *disjoint*:

- an entity can be a member of at most one of the subclasses of the specialization
- Specified by ***d*** in EER diagram

- If not disjoint, specialization is *overlapping*:
  - that is the same entity may be a member of more than one subclass of the specialization
- Specified by ***o*** in EER diagram

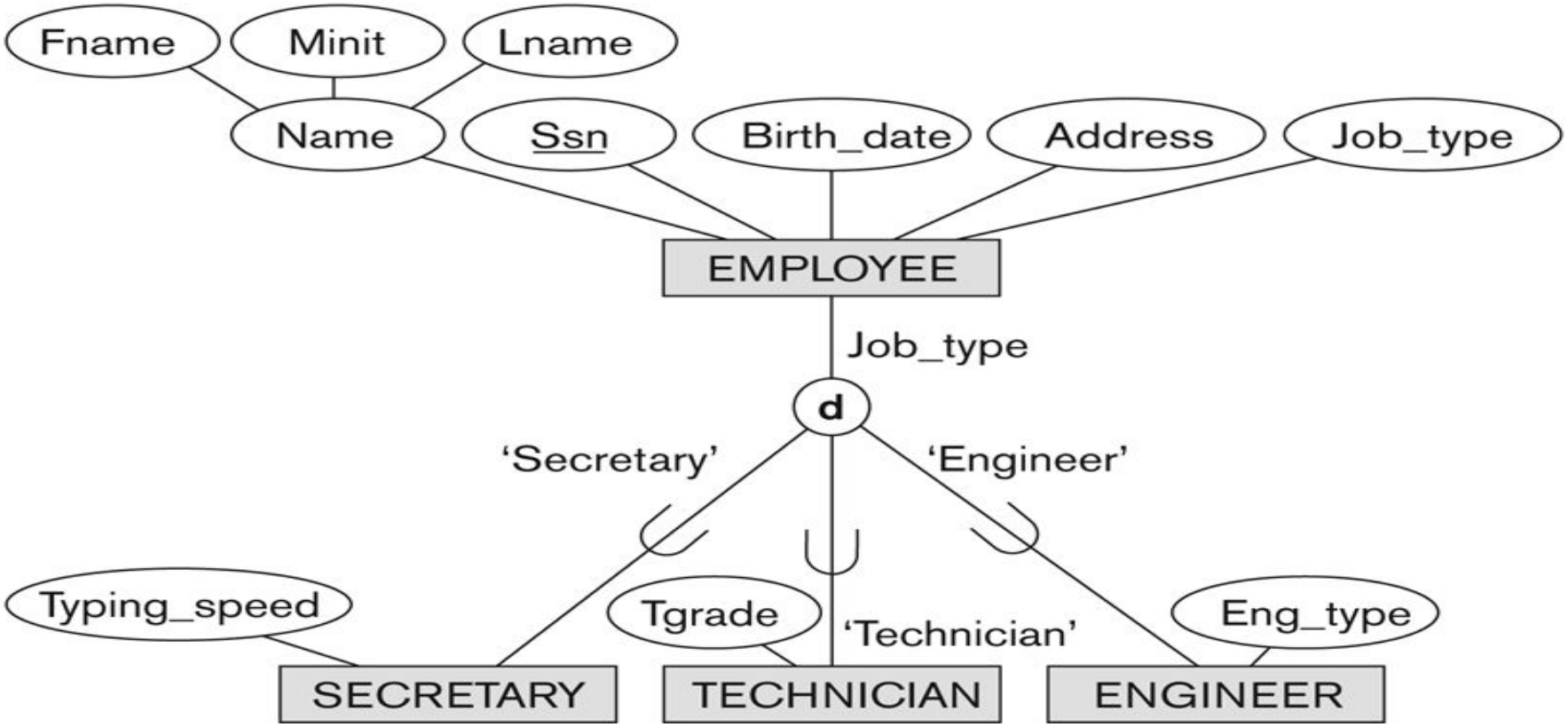
# Completeness Constraint:

- *Total* specifies that every entity in the superclass must be a member of some subclass in the specialization/generalization
- Shown in EER diagrams by a ***double line***

- *Partial* allows an entity not to belong to any of the subclasses
- Shown in EER diagrams by a single line

# Example of disjoint partial Specialization

**Figure 4.4**
EER diagram notation for an attribute-defined specialization on Job_type.




*Source: Fundamentals of database systems / Ramez Elmasri, Shamkant B. Navathe.—6th ed, Pearson Publications*

# Example of overlapping total Specialization

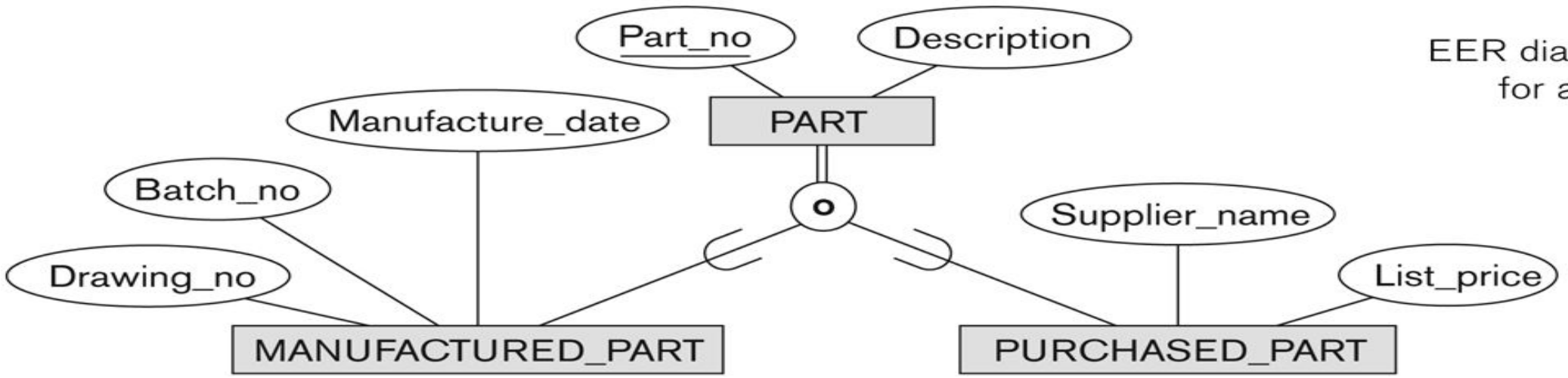



**Figure 4.5**
EER diagram notation for an overlapping (nondisjoint) specialization.

*Source: Fundamentals of database systems / Ramez Elmasri, Shamkant B. Navathe.—6th ed, Pearson Publications*

# Structures in Specialization

## Multiple Specializations

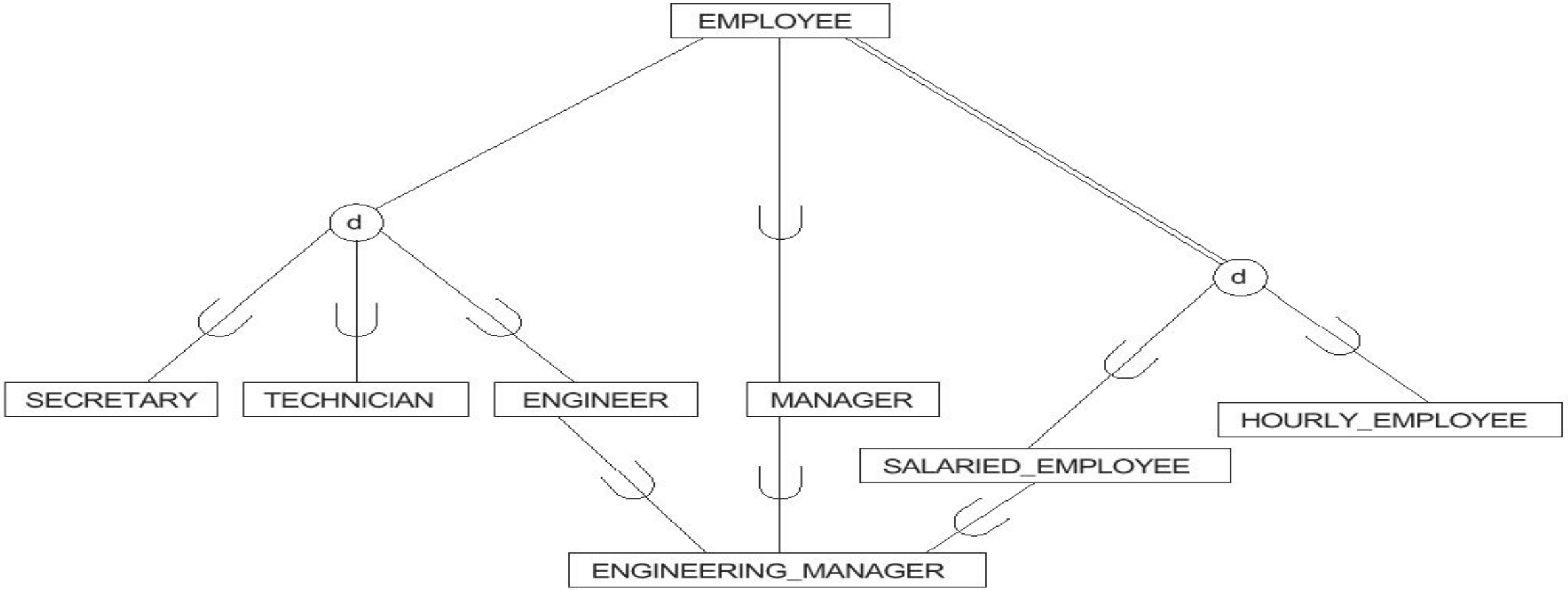
## Specialization Hierarchy

- ***Hierarchy*** has a constraint that every subclass has only one superclass (called ***single inheritance***); this is basically a ***tree structure***

## Lattice Specializations

- A subclass may belong to more than one class (called ***multiple inheritance***)

**Figure 4.6** A specialization lattice with the shared subclass ENGINEERING_MANAGER.




*Source: Fundamentals of database systems / Ramez Elmasri, Shamkant B. Navathe.—6th ed, Pearson Publications*

# Specialization / Generalization Lattice Example (UNIVERSITY)

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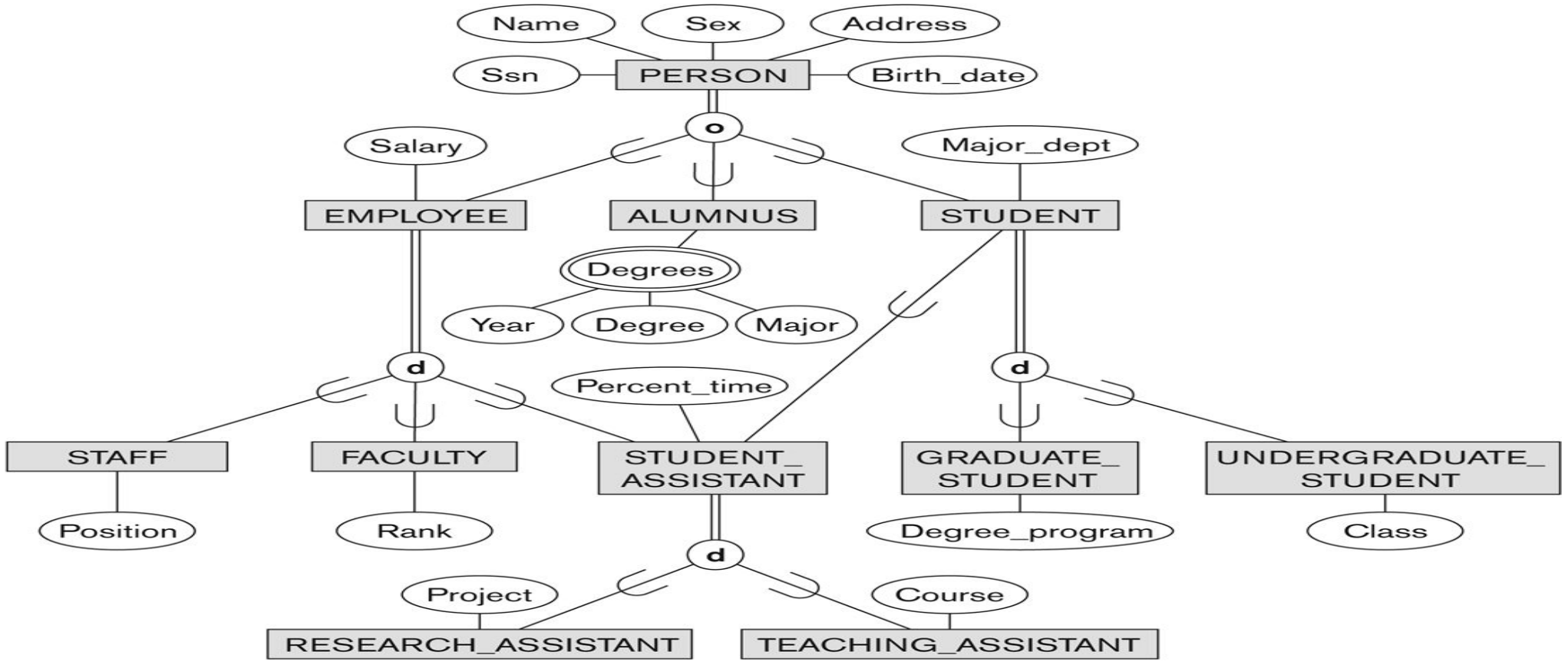
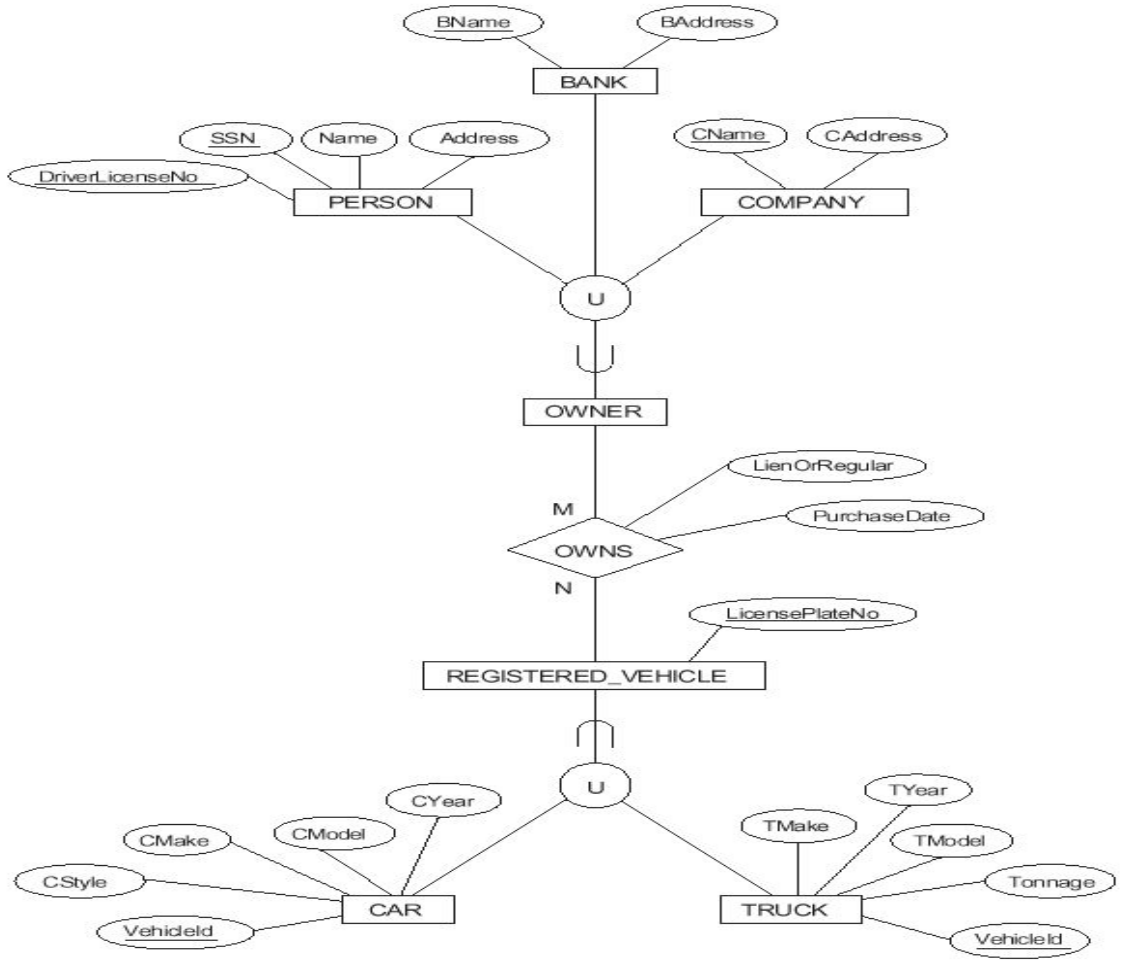



**Figure 4.7**
A specialization lattice with multiple inheritance for a UNIVERSITY database.

*Source: Fundamentals of database systems / Ramez Elmasri, Shamkant B. Navathe.—6th ed, Pearson Publications*

# UNION

**Figure 4.8** An illustration of how to represent the UNION of two or more entity types/classes using the category notation. Two categories are shown: OWNER and REGISTERED_VEHICLE.




*Source: Fundamentals of database systems / Ramez Elmasri, Shamkant B. Navathe.—6th ed, Pearson Publications*